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Information Framework (SID)

Catalog Business Entities

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Notice

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Direct inquiries to the TM Forum office:

240 Headquarters Plaza,
East Tower – 10th Floor,
Morristown, NJ 07960 USA
Tel No. +1 973 944 5100
Fax No. +1 973 944 5110
TM Forum Web Page: www.tmforum.org

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1 Catalog Business Entities

The Catalog Management Guidebook describes the usage of “Catalogs” to provide consistency of catalog model definitions across the Product, Service, and Resource domains, and to support the federation of catalogs and the ability to create a catalog containing any entity and/or its specification.

A generalized Catalog ABE contains generic (abstract) CatalogSpecification and Catalog entities used to define types of catalog specifications and types of catalog entities. The patterns used to model Catalog business entities are the EntitySpecification/Entity and the CharacteristicSpecification/CharacteristicValue patterns. It also uses relationship entities as an alternate to the Composite/Atomic pattern. It embraces the existing entity of ‘Product Catalog ABE’ definition present before this guidebook was developed.

1.1 Catalog Specification

Basic entities related to a CatalogSpecification and relationships with the existing entities are shown in the Figure Cat.01 – Catalog Specification.

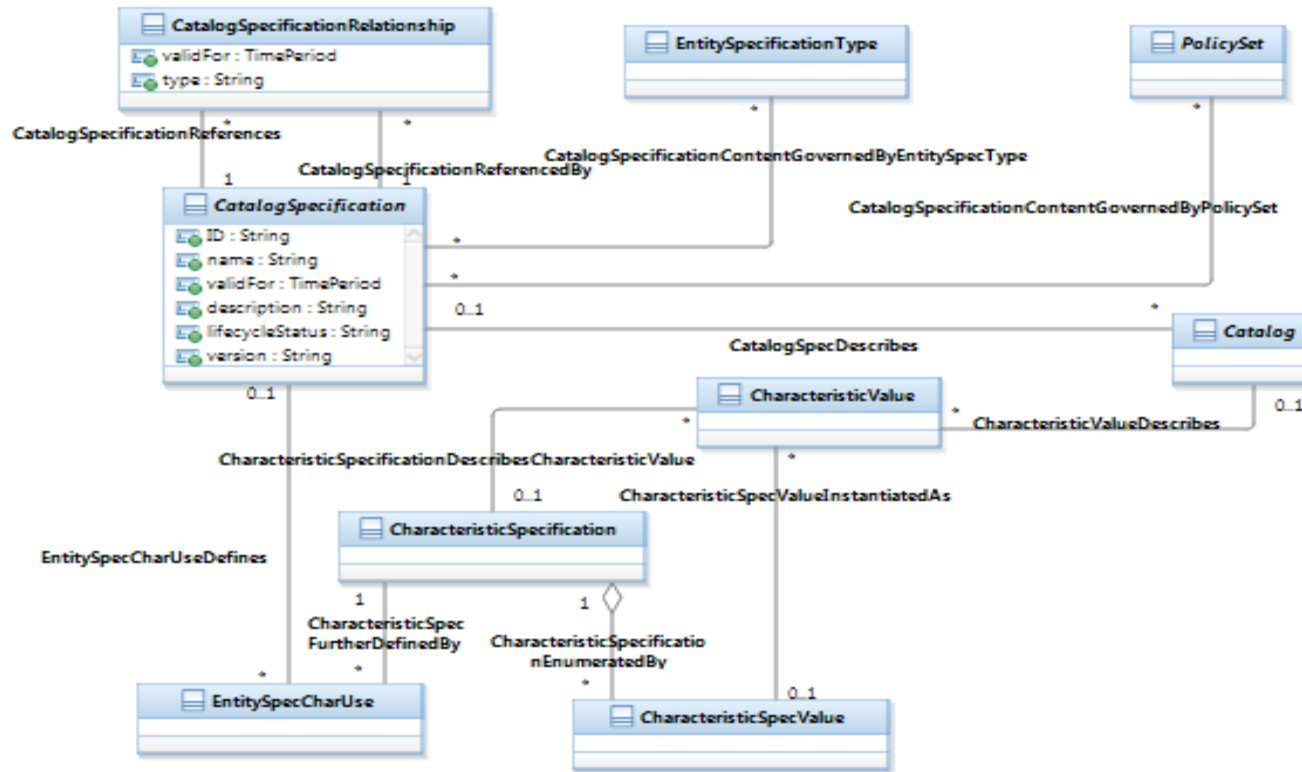


Figure Cat.01 – Catalog Specification

The CatalogSpecificationRelationship entity provides the ability to group specifications much like the Composite/Atomic pattern using the “type” attribute to specify that the relationship is a composite/atomic. For example, a Smart Home CatalogSpecification could be comprised of a Smart Home Security CatalogSpecification and a Smart Home Appliance CatalogSpecification. The “type” attribute can also be used to indicate that one CatalogSpecification supersedes another or is included with another.

The content of associated Catalogs can be dynamically defined using relationship to EntitySpecificationType for simple rules and/or to PolicySet for more complex rules. EntitySpecificationType is a way to create categories of any specification entity to which it or its subclasses are related, such as a ProductSpecificationType which is a subclass of EntitySpecificationType. For example, a Smart Home Security ProductSpecificationType could be related to the Smart Home Security CatalogSpecification. This relationship would enable a Smart Home Security Catalog, in this case a ProductCatalog, to be dynamically created. If there are more complex rules, such as limiting the catalog to only containing ProductOfferings that are in surplus status then a PolicySet could be defined to only include ProductOfferings that have this status.

The Characteristic pattern can be used to define dynamic attributes for any CatalogSpecification along with the attributes values for any Catalog.

1.2 Catalog

A CatalogSpecification defines a Catalog. The business entities that inherit from catalog: are ProductCatalog, ServiceCatalog and ResourceCatalog, EntityCatalog, and FederatedCatalog. EntityCatalog and FederatedCatalog will be described in a later section of this guide book.

Figure Cat.02 - Catalog Entities and Associations depicts the relationship of Catalog entities to the existing entities of information framework in the Product, Service and Resource domains.

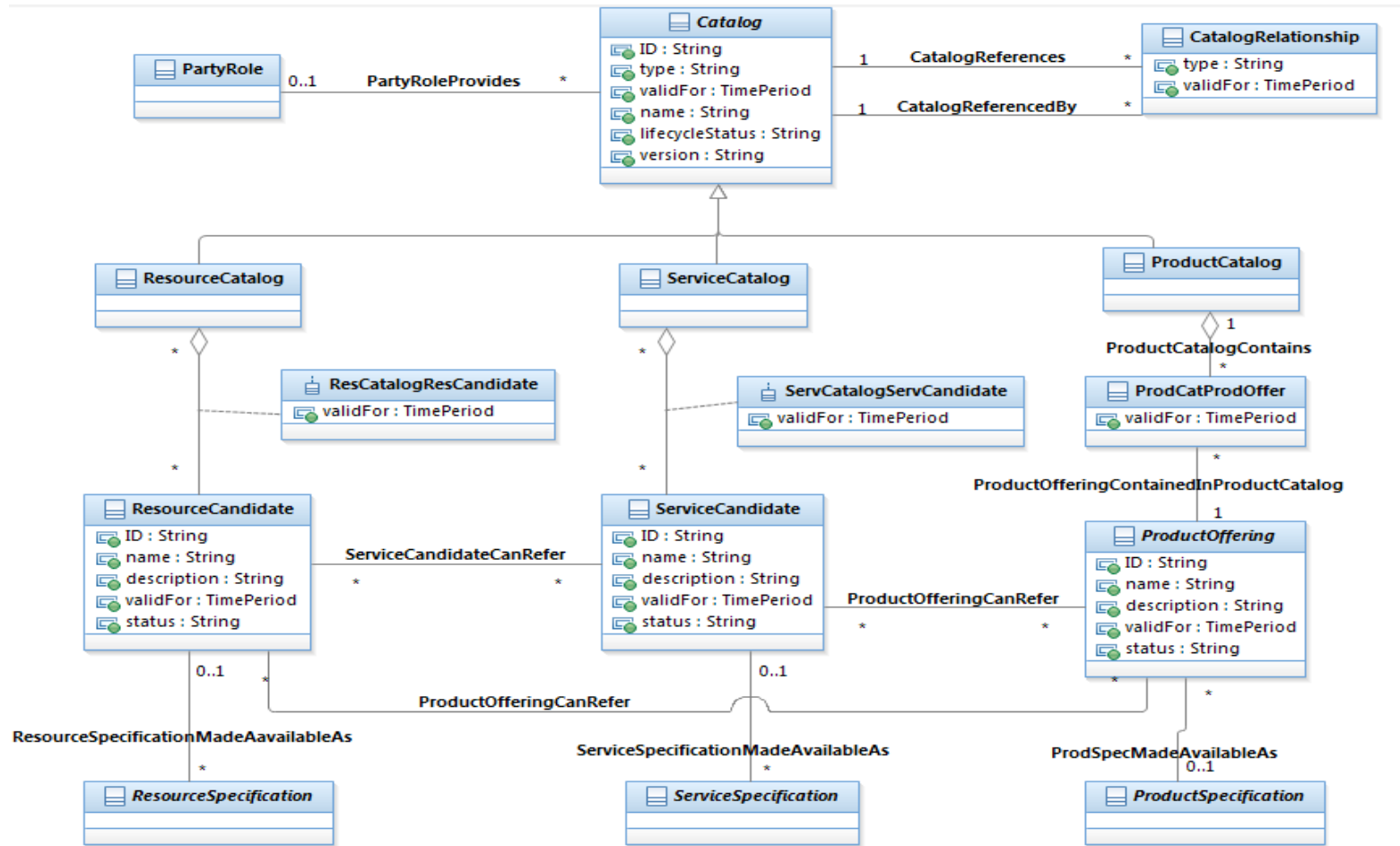


Figure Cat.02 – Catalog Entities and Associations

In this figure there are two more entities that are introduced through this guidebook. They are ServiceCandidate and ResourceCandidate. These two entities are defined to enable the respective specifications (ServiceSpecification and ResourceSpecification) to be available for inclusion in respective catalogs. As such the instances of “candidate” entities represent a subset of all the respective “specification” entities. For example, obsolete replacement ResourceSpecifications may not be of interest from a catalog perspective.

The PartyRole who provides the Catalog can also be specified. For example, a Product Catalog provided by an application developer.

1.3 Product/Product Catalog

A ProductCatalog is a collection of ProductOfferings, intended for a specific DistributionChannel, enhanced with additional information such as SLA parameters, invoicing and shipping details. As described in the Products addendum document, each ProductOffering in a ProductCatalog combines pricing and availability information with ProductSpecifications that describe the relationships between Products, the Services used to realize the Products, and the Resources they require.

An enterprise may choose to present its ProductOfferings to multiple audiences through multiple channels. Therefore, although the collection of all available ProductOfferings may be considered “the” ProductCatalog, the information model should support the definition and differentiation of multiple ProductCatalogs, containing possibly overlapping subsets of the ProductOffering universe.

1.3.1 Product Catalog and Product Catalog Specification

A ProductOffering may be “published” – made visible – in any number of ProductCatalogs, or in none. The most important association in the Product Catalog ABE is the publication of a ProductOffering in a ProductCatalog. Attributes affecting that association, such as the time period during which a ProductOffering is valid in a particular ProductCatalog, are contained in the association class ProdCatalogProdOffer.

The ProductCatalog itself may have attributes that define its identity or existence, such as type (String) and validFor (TimePeriod). But most of the variant data related to a ProductCatalog is controlled by the ProductCatalogSpecification entity.

Key associations with entities in other domains include a ProductCatalog’s availability through a SalesChannel, and the publication of a ProductCatalog in a GeographicArea. The availability of a ProductCatalog through a specific SalesChannel implies that each ProductOfferingPrice that values a ProductOffering appearing in that ProductCatalog must also be made available through the same SalesChannel. In turn, each ProductOfferingPrice may be advertised in any number of ProdCatalogProdOffers.

ProdCatalogProdOffer may contain additional information regarding the offering in the catalog, such as the location of an offering in a catalog. These basic entities are shown in Figure Cat.03 – Product Catalog Definition and Basic Relationships.

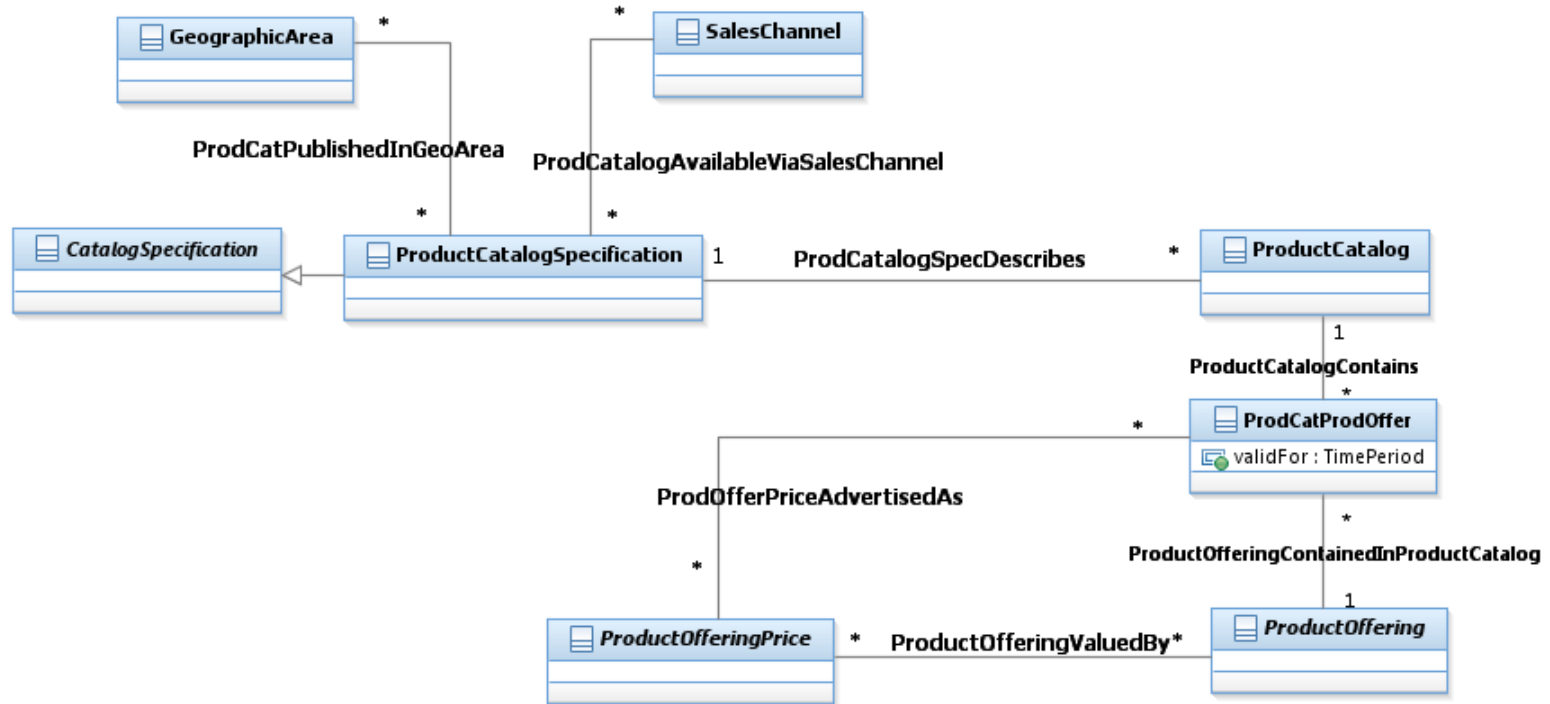
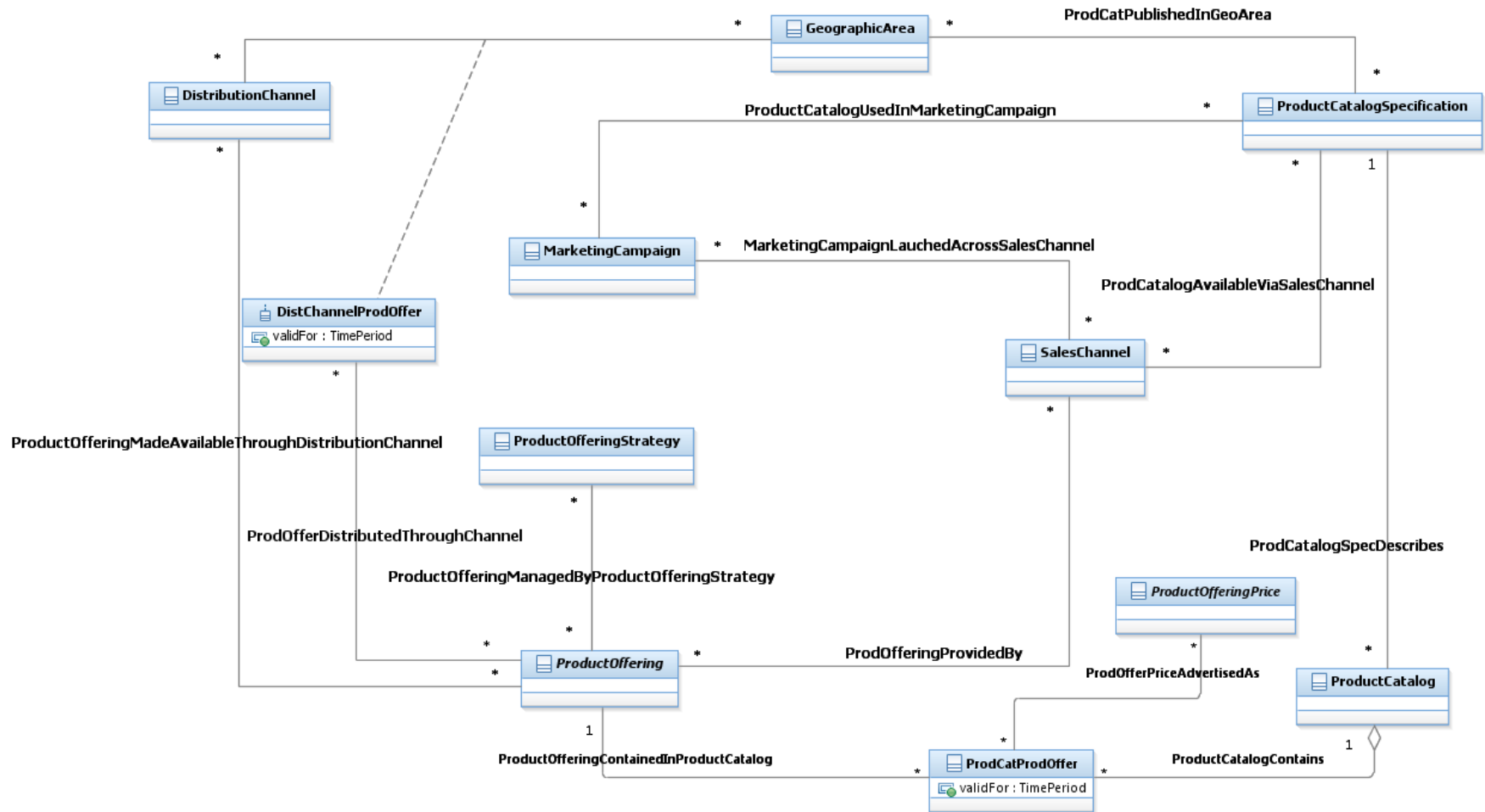


Figure Cat.03 – Product Catalog Definition and Basic Relationships

1.3.2 Related Entities

Several additional entities complete the Product Catalog. A DistributionChannel is responsible for providing ProductOfferings in a specific GeographicArea. Each ProductOffering is managed under a ProductOfferingStrategy that supports any number of MarketingCampaigns. MarketingCampaigns are run through SalesChannels, and use ProductCatalogs selected by ProductCatalogSpecification. These entities are shown in Figure Cat.04 – Product Catalog Entities and Associations.



1.4 Service Catalog

A ServiceCatalog is a group of ServiceSpecifications made available through ServiceCandidates that an organization provides to the consumers (internal consumers like its employees or B2B customers or B2C customers). A ServiceCatalog typically includes name, description and time period that is valid for. It will have a list of ServiceCandidates. A ServiceCandidate is an entity that makes a ServiceSpecification available to a catalog. A ServiceCandidate and its associated ServiceSpecification may be “published” – made visible – in any number of ServiceCatalogs, or in none. One ServiceSpecification can be composed of other ServiceSpecifications. These ServiceSpecifications may also be published. Attributes affecting that association, such as the time period during which a ServiceCandidate is valid in a particular ServiceCatalog, are contained in the association class ServCatalogServCandidate. These basic entities are shown in Figure Cat.05 – Service Catalog Entities and Associations.

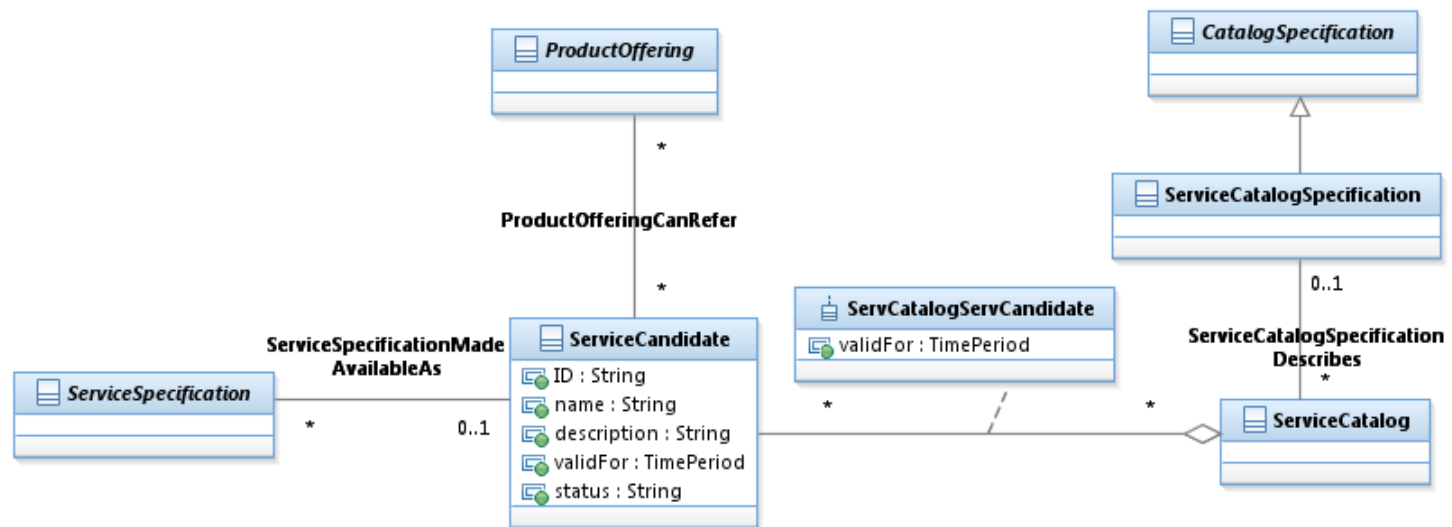


Figure Cat.05 – Service Catalog Entities and Associations

1.5 Resource Catalog

A ResourceCatalog is a list of ResourceSpecifications made available through ResourceCandidates that an organization will provide it to its consumers (internal or external). Not all ResourceCandidates are made available to a consumer. A ResourceCatalog typically includes name, description and time period that it is valid for. It will have a list of ResourceCandidates. A ResourceCandidate is an entity that enables the ResourceSpecifications to a catalog. A ResourceCandidate and its ResourceSpecification may be “published” – made visible – in any number of ResourceCatalogs, or in none. One ResourceSpecification can be composed of other ResourceSpecifications. These ResourceSpecifications may also be published. Attributes affecting that association, such as the time period during which a ResourceCandidate is valid in a particular ResourceCatalog, are contained in the association class ResCatalogResCandidate. These basic entities are shown in Figure Cat.06 – Resource Catalog Entities and Associations.

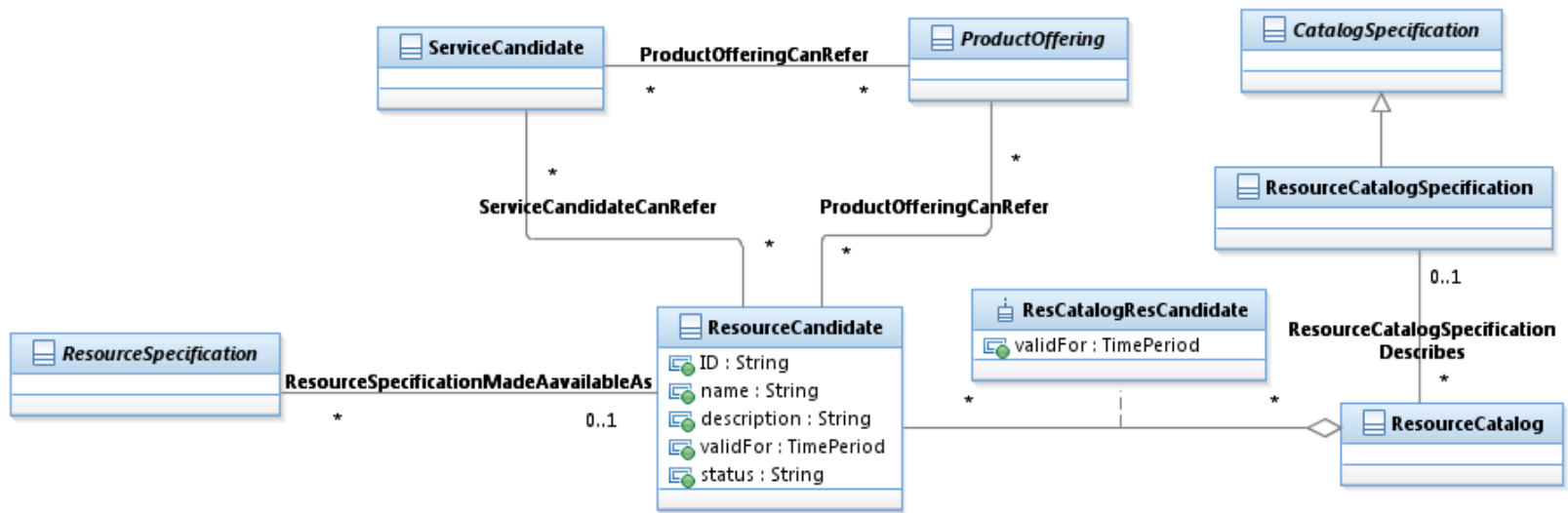


Figure Cat.06 – Resource Catalog Entities and Associations

1.6 Entity Catalog and Federated Catalog

The advent of network function virtualization and other innovations in the realm of virtualization presented some challenges for the specialization of the CatalogSpecification and Catalog entities into ones for Product, Service, and Resource. From these and other requirements arose the need to introduce catalogs that could contain any specification and any entity and combine (federate) them in any way required and to be able to create them dynamically at run time. Support for this was explained in section 1.1 Catalog Specification, along with some examples.

The virtualization requirements for catalogs in this section were extracted from TR244 - TM Forum Information Framework Enhancements to Support ZOOM. Minor modifications made to them to remove references to proposed ZOOM catalog model.

1.6.1 Examples

The following two examples illustrate these requirements.

1.6.1.1 Creating a VNFCatalog

A VNFCatalog is a fundamental requirement of ETSI NFV compliance. However, the ETSI NFV definition of a VNF Catalog is not specific, so vendor and provider implementations vary widely. Specifically:

- VNF is not always offered as a Product
- A VNF typically consumes Resources, but does not have to expose them
- A VNF typically produces Services, but does not have to expose them.

Therefore, a VNFCatalog can be something else than a ProductCatalog, ResourceCatalog or ServiceCatalog.

1.6.1.2 Dynamic NFV Catalog Federation

The capability to define and federate dynamically generic catalogs containing items belonging to both Service and Resource domains is illustrated in the Dynamic NFV Catalog Federation User Stories developed in ZOOM. Additional examples can be found in the Digital Services Reference Architecture Guide / Catalog Federation Direction tab.

1.6.2 Entity and Federated Catalog - Specifications and Entities

Figure EC.01 - Entity Catalog Specification show the addition of a FederatedCatalogSpecification and an EntityCatalogSpecification as subclasses of CatalogSpecification. Like the other types of CatalogSpecifications their content can be dynamically defined at run time. The EntityCatalogSpecification enables any catalog, such as the VNFCatalog mentioned in 1.6.1 Examples, to be defined. The FederatedCatalogSpecificationComprisedOf association enables any combination of the subclasses to be federated supporting the dynamic NFV Catalog Federation requirement in 1.6.1 Examples.

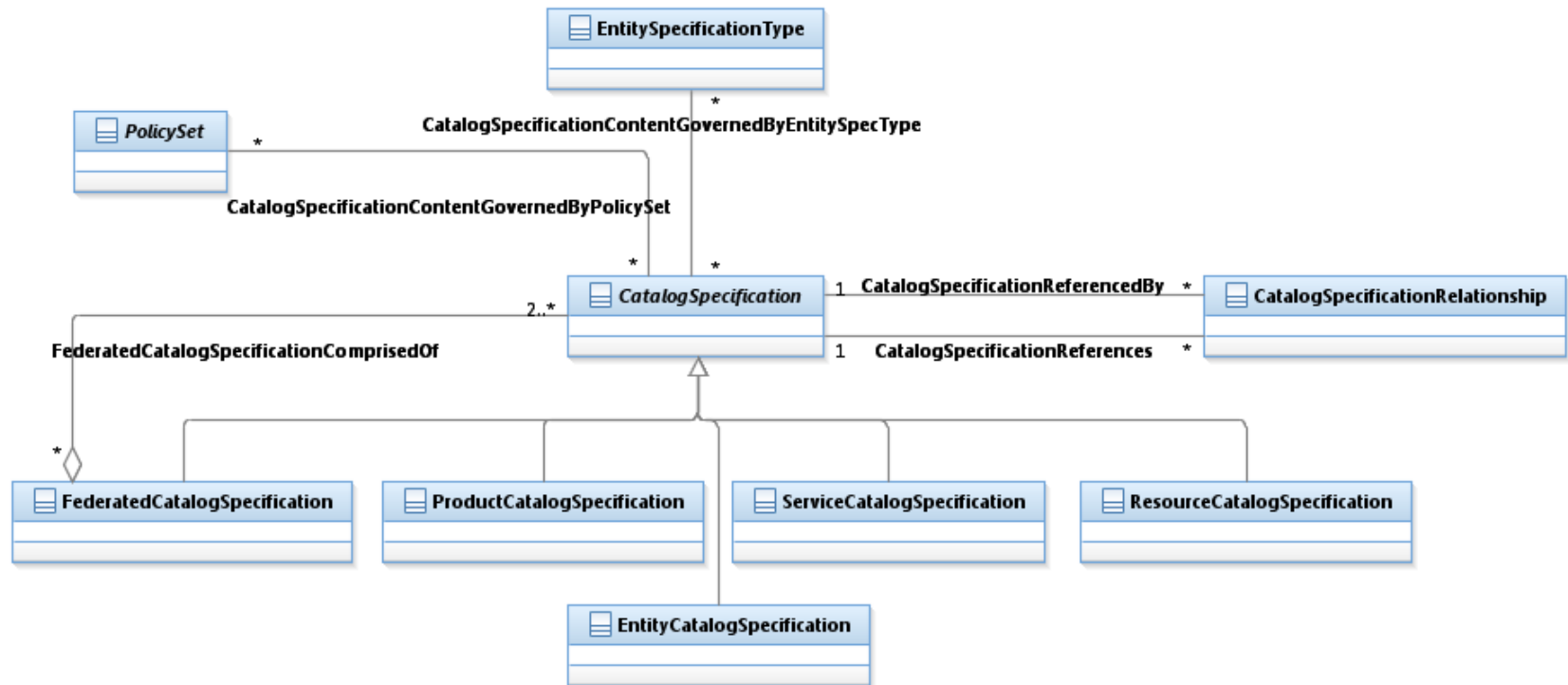


Figure EC.01 - Entity Catalog Specification

Figure EC.02 - Entity Catalog shows the Catalog and its subclasses that can be dynamically generated at run time based on their specification.

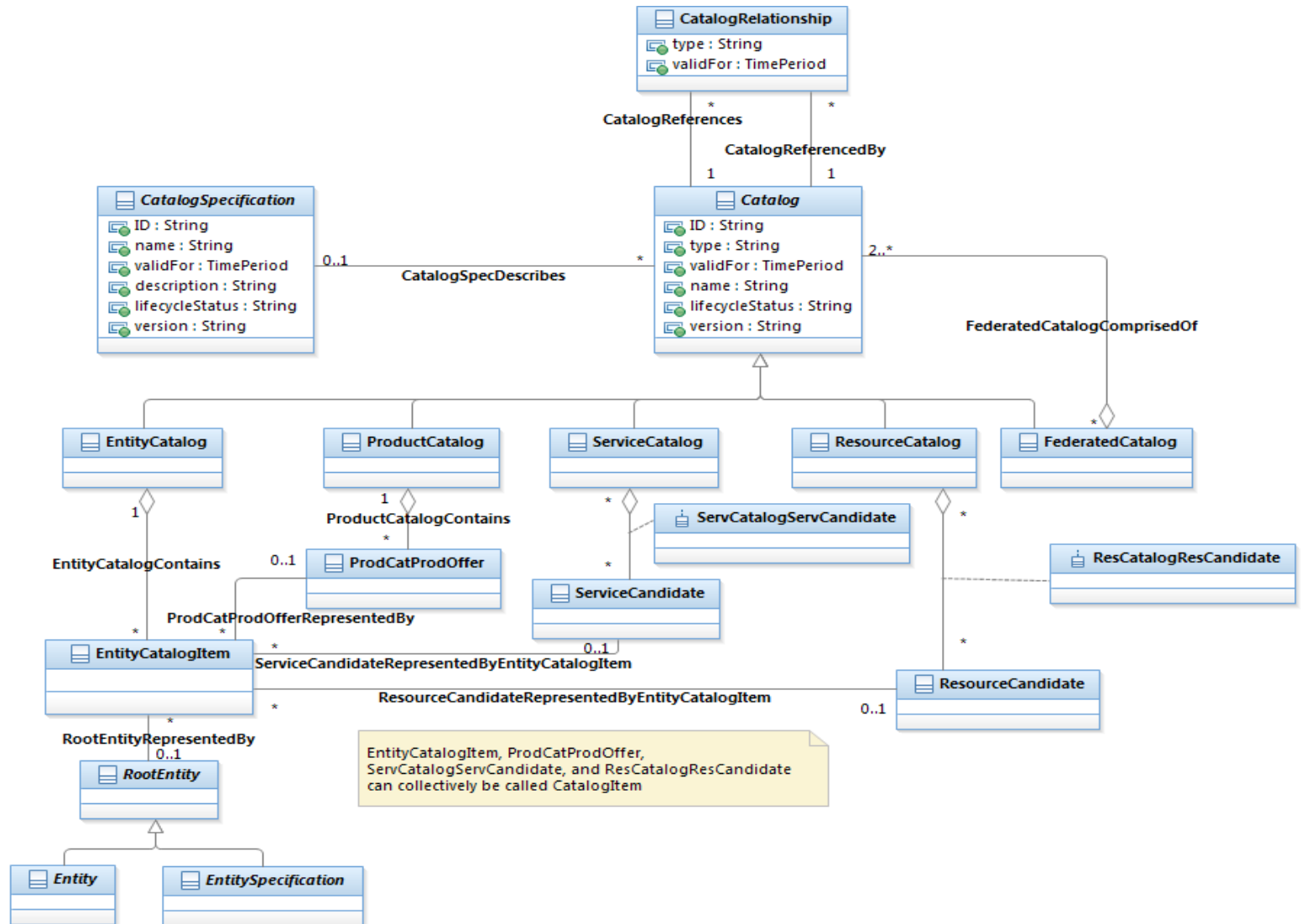


Figure EC.02 - Entity Catalog

An EntityCatalogItem can be one that is contained in a Product, Service, or Resource Catalog, or any Entity or EntitySpecification that has been included in the definition of its specification using the associations to EntitySpecificationType and/or PolicySet.

1.7 Example - Generating Dynamic Catalogs

The following example shows how a Broadband catalog can be specified and dynamically generated.

Originally, vendor X's catalog Broadband catalog only contained a physical gateway and phones for VoIP. But now the vendor wants to build a Broadband Procurement Catalog that contains any broadband offering that can be acquired.

To be able to dynamically build / update the Catalog rules related to the corresponding CatalogSpecification are needed.

The rules specify:

- When the Catalog has to be updated by creating a triggering Event and relating it to the specification for the Broadband Procurement Catalog
- Which "category", or ProductSpecificationType, and its related ProductSpecifications are to have their ProductOfferings, is to be included in the catalog, by relating them to the specification for the Broadband Procurement Catalog.

Figure EC.03 – Broadband Procurement Catalog shows the CatalogSpecification instance and Catalog instances and instances of their related entities. According to the rule specified for the Broadband Catalog Specification, a Broadband Procurement Catalog is first built in January; then another one is built in February. It contains all that can be acquired for a Virtualized Gateway, in addition to the Physical Gateway and the IP Phone. The example also illustrates that the “IP Phone” instance of ProductSpecification is related to two instances of ProductSpecificationType and therefore the corresponding ProductOffering is available for inclusion in other catalogs.

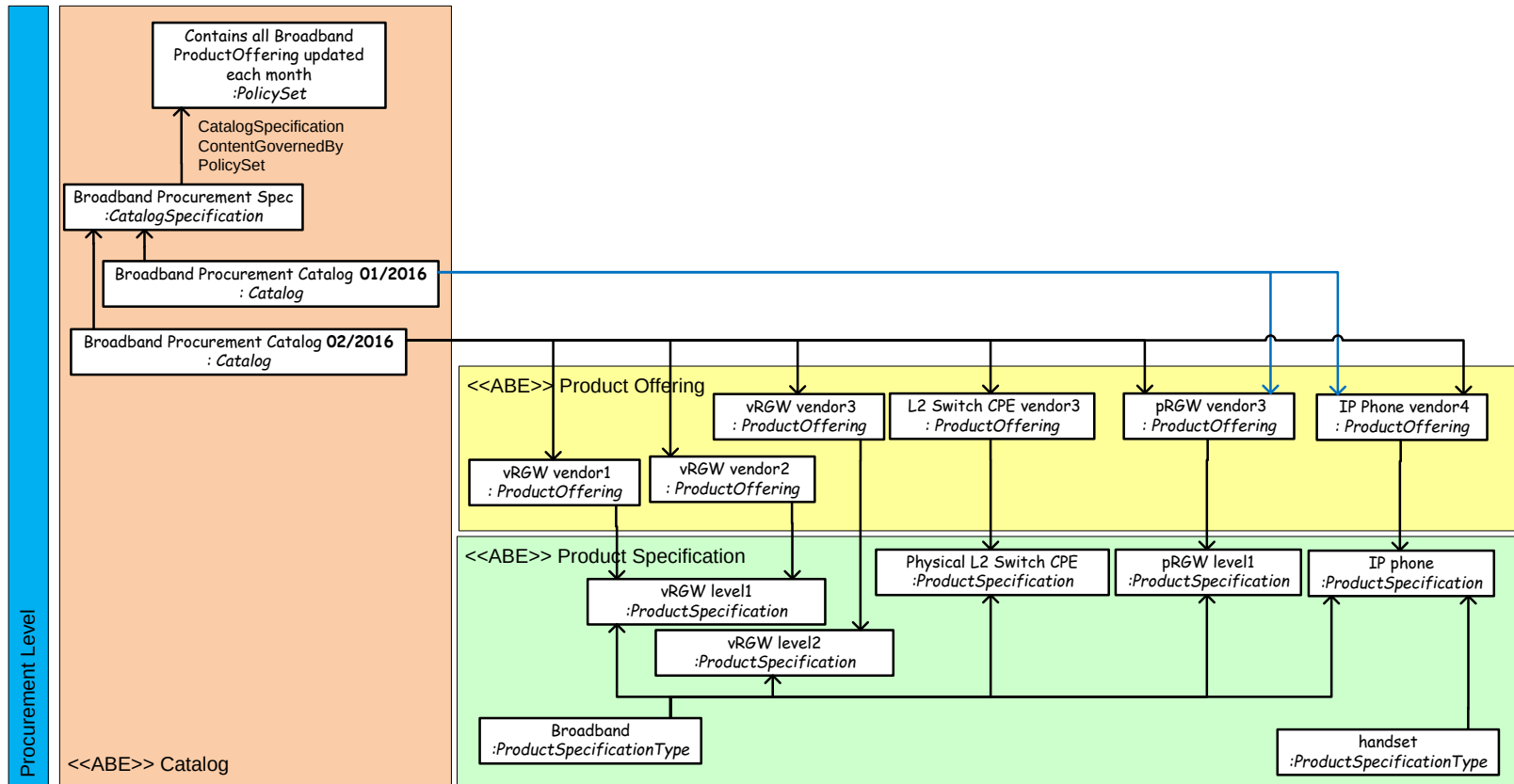


Figure EC.03 – Broadband Procurement Catalog

2 Administrative Appendix

This Appendix provides additional background material about this document.

2.1. Document History

2.1.1 Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1	14/08/2012	Raj Ravi	first issue of TM Forum contribution
0.2	28/09/2012	Raj Ravi, John Reilly and Elaine Maher	Consolidation of the definitions and documentation
0.5	28/09/2012	Catalog Management project team	Final edits of definitions
1.0	02/10/2012	Raj Ravi	Added the generalization association between CatalogSpecification to ProductCatalogSpecification (Cat 04), ResourceCatalogSpecification (Cat05) & ServiceCatalogSpecification (Cat06). Removed the 'type' attribute that was present in the earlier version in CatalogSpecRelationship Added the entity definition for CatalogSpecRelationship (Section 1.6.2). Removed the attribute list from the ProductCatalog

Version Number	Date Modified	Modified by:	Description of changes
			definition in the entity definition (Section 1.6.3.2 in the previous version.) Updated the 'Company Contact Details' section with the team list and also removed the email and phone contact details. Added the ProductCatalogSpecification, ResourceCatalogSpecification and ServiceCatalogSpecification entities in the entity definitions section.
1.1	25/10/2012	Alicja Kawecki	Minor formatting/cosmetic corrections prior to web posting and Member Evaluation
1.1.1	7/11/2013	Alicja Kawecki	Applied rebranding, updated cover, header, footer & Notice
2.0	16 April 2014	Avi Talmor	Copy of all pertinent information to the model and removal of Business Entity Definition section
2.0.1	24 April 2014	Alicja Kawecki	Minor formatting edits prior to posting for Member Evaluation
2.0.2	25 Sep 2014	Alicja Kawecki	Updated cover and Notice to reflect TM Forum Approved status
15.0.0	1 May 2015	John Reilly	Updated figures and text to show association between PartyRole and Catalog. Changed ProdCatProdOffer from an association class to a class.

Version Number	Date Modified	Modified by:	Description of changes
15.0.1	19 May 2015	Alicja Kawecki	Updated cover, header, Notice
15.0.2	19 Nov 2015	Alicja Kawecki	Updated cover and Notice to reflect TM Forum Approved status
16.0.0	10 May 2016	John Reilly	Updated to support dynamically defined and generated catalogs, including VFNCatalogs and dynamic NFV catalog federation
16.0.1	25 May 2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16
16.5.0	31 Oct 2016	Kamal Maghsoudlou	Updated figures to demonstrate new attributes
16.5.1	18 Nov 2016	Alicja Kawecki	Minor cosmetic corrections prior to publication for Fx16.5
16.5.2	16 Feb 2017	Alicja Kawecki	Updated cover, header, footer and Notice to reflect TM Forum Approved status; applied rebranding

2.1.2 Release History

Release Number	Date Modified	Modified by:	Description of changes
16.5.0	31 Oct 2016	Kamal Maghsoudlou	Updated figures to demonstrate new attributes

2.2. Acknowledgments

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- Plamen Stanoev, Ericsson – **Contributor**
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- Yuval Stein, TEOCO – **Contributor and Reviewer**
- Kamal Maghsoudlou, Ericsson – **Contributor and Reviewer**